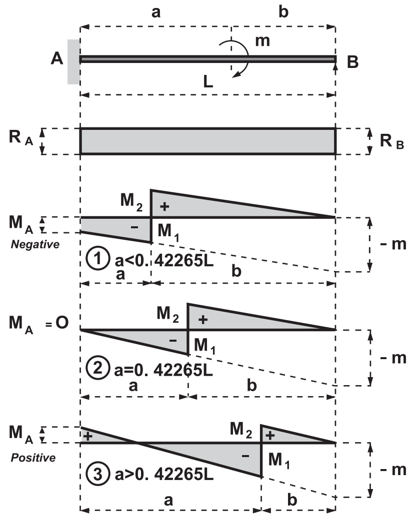
I-BEAMS

Propped cantilever — Moment applied at any point



Span = L

Applied moment = m

$M_A = \frac{L^2-3b^2}{2L^2}m$ $\qquad i_B = \frac{ma}{4EIL}(2b-a)$

$R_A = -R_B = -\frac{3(L^2-b^2)}{2L^3}m = -\frac{m+M_A}{L}$

$M_1 = -\frac{m}{L^3}(a^3+\frac{3}{2}a^2b+b^3)$

$M_2 = \frac{3mab}{L^3}(b+\frac{a}{2}) = m+M_1$

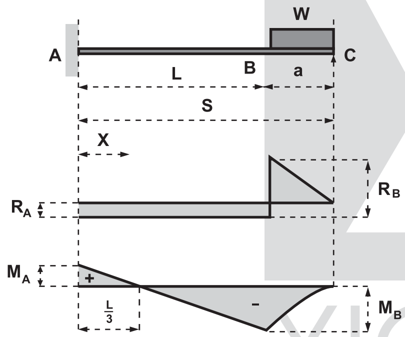
$M_1 = -0.42265m$

$M_2 = 0.57735m$

$M_1 =$

$M_2 =$ as for ①

Propped cantilever — Uniform load on length beyond prop



Span = L $\qquad$ Full length = S

Uniform load = W

$R_A = -\frac{3Wa}{4L}$ $\qquad R_B = \frac{W}{L}(S-\frac{a}{4})$

$M_A = \frac{Wa}{4}$ $\qquad M_B = -\frac{Wa}{2}$

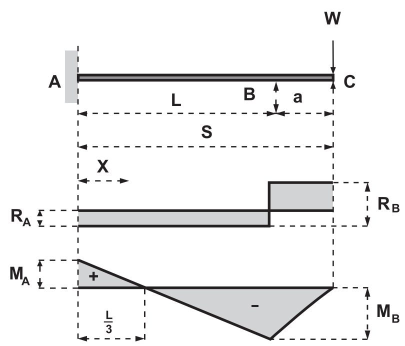
Deflection at C = $\delta_{max} = \frac{Wa^2S}{8EI}$

Max. negative deflection at X = $\frac{2}{3}L$ $\qquad \delta_{neg} = -\frac{WL^2a}{54EI}$

Slope at C = $i_C = \frac{Wa}{8EI}(S+\frac{a}{3})$

at X = $\frac{L}{3}$ from A, M = 0

Propped cantilever — Point load at free end



Span = L $\qquad$ Full length = S

Point load = W

$R_A = -\frac{3Wa}{2L}$ $\qquad R_B = \frac{W}{L}(S+\frac{a}{2})$

$M_A = \frac{Wa}{2}$ $\qquad M_B = -Wa$

Deflection at C = $\delta_{max} = \frac{Wa^2}{4EI}(S+\frac{a}{3})$

Max. negative deflection at X = $\frac{2}{3}L$ $\qquad \delta_{neg} = -\frac{WL^2a}{27EI}$

Slope at C = $i_C = \frac{Wa}{4EI}(S+a)$

at X = $\frac{L}{3}$ from A, M = 0